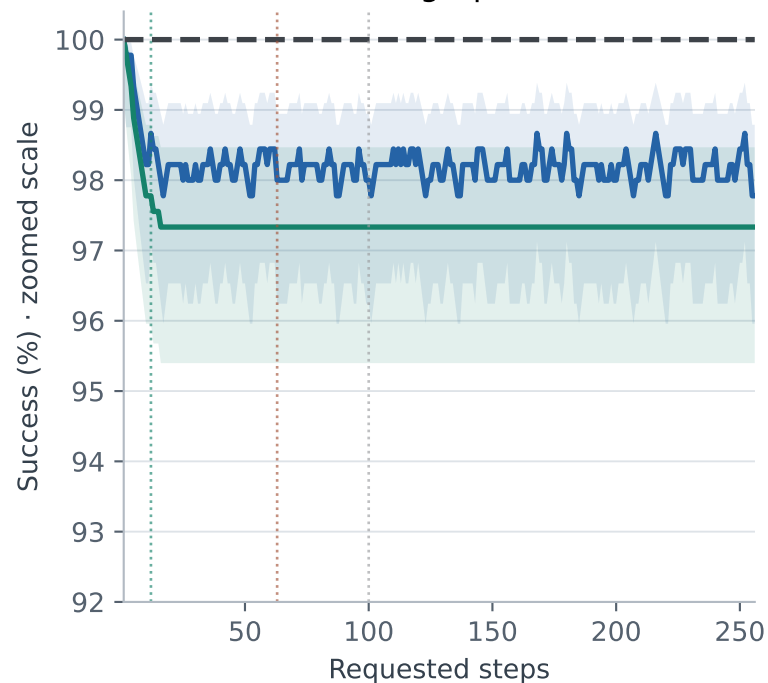


Graph structure and depth generalization

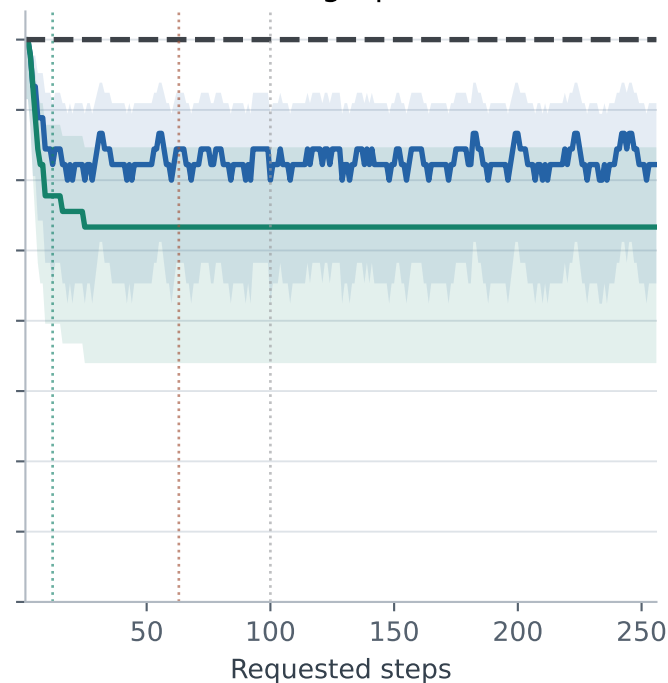
1,350 independent graphs · 345,600 paired queries · every count 1-256

Correct answer + exact stopping loop Every intermediate step correct Exact stopping loop

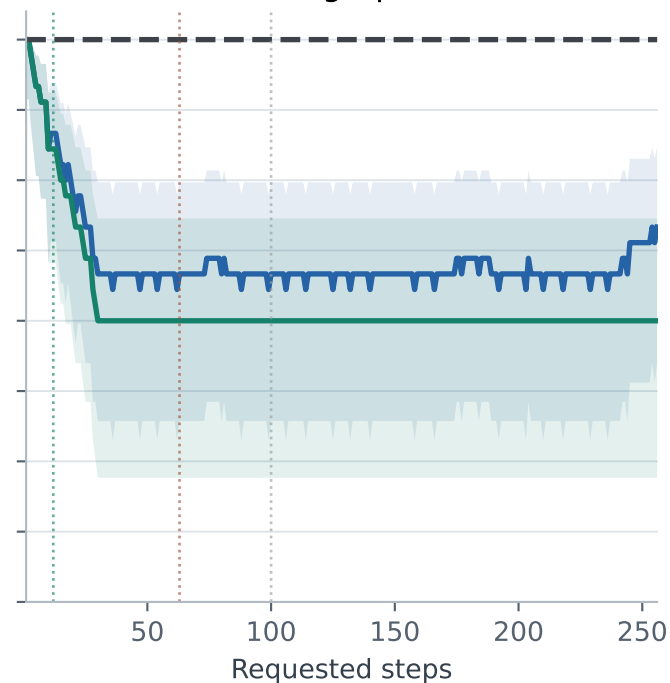
Random functions
450 graphs



Permutations
450 graphs



Full 26-state cycles
450 graphs



Dotted lines: R training unroll 12 · largest trained count 63 · three-digit boundary 100.

Shading: pointwise 95% graph-level Wilson intervals; repeated horizons are paired, not independent.